

Uncovering Themes in Junk Science Articles with BERTopic

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Abstract

This study examines how junk science articles are written and what themes they contain using topic modeling on Fake News Corpus dataset. Junk Science spreads misleading claims online and can weaken trust in real science and research. The project extracted all articles labeled “junksci” from Fake News Corpus dataset. It was cleaned and text was lemmatized and used BERTopic on 17,408 articles to group them into 26 topics, including one outlier topic. The remaining 25 topics were interpreted and labeled using their top keywords and mapped to standard branches of science. These labeled topics provide a clear overview of common junk science themes that can support future work on misinformation detection and clearer public communication.

Keywords: junk science, topic modeling, fake news, BERTopic, misinformation

Introduction

Misinformation and junk science spread extensively on many digital platforms such as social media, blogs, articles, and other online venues. For example, a social media post from blowingfact claims “scientists in Japan can now erase memories like in a movie” while using a poster from a popular memory-loss movie to suggest that the lab research lets people delete real-life memories. Even though the underlying research only shows that researchers were able to cause partial memory disruption in mice using specialized lab techniques, the post was turning

complex, early state animal studies into very bold, strong, and dramatic promises for people and hiding real science behind. Similarly, other junk science articles and posts that are present misleading and wrong scientific claims without any proper evidence or peer review, which can misrepresent public understanding of science and destroy trust in correct scientific information. This problem has increased with the speed and ease of how information is shared, making it harder for people to discern credible sources from deceptive ones.

Previous research has used Fake News Corpus (Fake News Corpus, n.d., <https://github.com/several27/FakeNewsCorpus>), an opensource dataset of news articles that is labeled by source type, to train machine learning models that classify news as real or fake, and these models have performed very well at detecting fake news, with recent misinformation detection studies reporting that almost half of evaluated models reach accuracies of 0.95 or higher and many other fall are around 0.90-0.94 range (Chung & Chen 2025). However, they do not really look closely at junk science articles as a specific type of misinformation or examine their themes/main topics or their writing that makes these articles appear believable.

The purpose of this study is to analyze all the junk science articles from Fake News Corpus dataset to identify common topics using BERTopic. By organizing the data into clear categories using topic modeling, this work aims to: (1) discover the topical coverage of the dataset in junk science literature; and (2) identify possible challenges of using this dataset on misinformation detection.

This study is guided by the following question: What are the dominant junk science topics in the Fake News Corpus?

Related Literature

The Fake News Corpus dataset (Fake News Corpus, n.d., <https://github.com/several27/FakeNewsCorpus>), as introduced earlier, has been widely used in research and projects that are focused on fake news detection and misinformation analysis. For example, AIR Legend's GitHub repository has experiments using deep learning methods trained on the Fake News Corpus to classify news as fake or real (AIRLegend, 2025).

Other studies including a master's thesis on content based fake news detection and a journal article that combines BERT with LightGBM, have also used the Fake News Corpus to build and evaluate fake news detection systems and reported a very strong performance (Arnfield, 2023; Essa, Omar, & Alqahtani, 2023). This dataset helps researchers create tools that can better spot fake information online and also improve misinformation detection and management.

In misinformation research, several topic modeling methods have been used to study different patterns in news and social media. Earlier studies often used chance based or in other words probabilistic models like Latent Dirichlet Allocation (LDA) and matrix-based methods like Non-negative Matrix Factorization (NMF) to group articles into topics by looking at which words frequently appear together (Nath, 2020). More recent research uses neural or embedding-based models, including transformer-based approaches such as BERTopic, which turns each document into a numeric vector and can capture a more detailed meaning connections between words and articles that have noisy text.

Methodology

The Data and Data Extraction

The data for this research study comes from Fake News Corpus. It was created by developer Maciej Szpakowski (“several27”) in 2018 by scraping news articles from a curated list of 1,001 online domains collected on fact checking site OpenSources.co (Szpakowski, 2018). The public release includes about 9.4 million articles from 745 of these domains, that covers source types such as fake news, satire, extreme bias, conspiracy, junk science, hate news, clickbait, unreliable, political, and credible. This dataset was specially designed for training and evaluating fake news detection models. Each article is stored in a CSV file with different fields: id, domain, URL, title, full article content, metadata, and time stamps, and the dataset is mainly designed for training fake news detection models.

In this corpus, every article is labeled with a “type” based on the source domain, using the categories listed above. For this project, only articles with the domain type “junksci” (junk science) are used. This subset allows the study to focus on how junk science content is structured and presented in a larger misinformation environment.

In this study, junk science means information that looks like real science but does not follow good scientific rules. It often uses scientific words and few facts but ignores strong evidence or careful methods that can mislead people about what scientists actually know.

The Fake News Corpus release (approximately 30 GB) was downloaded and has extracted only the articles whose type of label was “junksci”. The junk science labeled articles were then loaded into a pandas DataFrame and then saved as a separate CSV file to create a focused subset for a deeper analysis.

Using the junk science subset, it was preprocessed by selecting only the content column, which contained the main article's text, and all text was converted to lowercase. Then, the text was cleaned by removing punctuation, non-alphabetic characters, and extra whitespace, and by deleting very short or empty documents. Next, common English stop words was removed and a WordNet lemmatizer was added to reduce words to their base form.

In addition, a custom stopwords list was built iteratively. After each initial BERTopic run, frequently appearing but uninformative words were manually inspected and added to the custom stopwords list. The model was refitted several times until the resulting topics contained fewer generic words and more meaningful keywords. In total, the final custom list contained 1583 custom stopwords beyond the standard English list, and this list was stored directly in the Jupyter Notebook as a Python list so it could be reused for later runs for the model.

Topic Modeling

Topic modeling was conducted using BERTopic, a neural topic modeling technique that combines transformer-based document embeddings with a class-based TF-IDF procedure to create interpretable topics (Grootendorst, 2021). After preprocessing, the cleaned text from each article's content field was stored as a separate string in Python list called documents, so that each element of documents represented one article to be passed into BERTopic. As a result, 17,408 documents were generated.

The BERTopic was trained on this list of junk science documents using the BERTopic pipeline, with the initial number of topics set to 51, where topic -1 is treated as an outlier topic that groups together documents that do not clearly belong into any coherent topic. To make sure that the themes are more compact and interpretable, the number of topics was then reduced from 51 to 26.

Moving on, the model embedded each document, reduced dimensionality, and then clustered documents into topics, returning a topic ID with a set of top keywords for each topic including an outlier topic –1. The number of top keywords for each topic was kept at a default value of 10 rather than increased, because using a larger number would introduce more generic, high-frequency words which would be less representative of each topic.

The models' output was stored in a DataFrame, a new column was added to summarize the interpreted themes based on the set of key words. These themes were added manually by mapping each topic to a corresponding branch of science using the Wikipedia “Index of branches of science” list.

Results

The final dataset used for topic modeling contained 17,408 cleaned junk science articles, each represented as one document in the model. Using BERTopic, 26 topics were generated in total, including an outlier topic labeled as -1 that grouped documents which did not fit well into any main cluster. For each of the remaining topic ID's (0-24), the model produced top keywords, which were stored in a DataFrame along with the topic numbers. Using the key words list, I manually interpreted every topic based on the top words and added a new column with short, specific labels.

Table 1. Topics &. labels of Junk Science articles from Fake News Corpus

Topic	Top 10 Words	Topic Name
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-1	youre, processed, mercury, level, effective, program, community, social, mainstream, meal	Outliers
0	glyphosate, vaccination, magnesium, effective, disorder, safe, yoga, protect, mercury, program	Toxicology
1	gods, design, core, present, date, deep, cave, heaven, creator, dating	Theology
2	bica, matt, paul, carbonnyc, browne, johannes, various, upton, mae, holden	Bromatology
3	criminal, debt, atmosphere, storm, security, atmospheric, financial, terrorist, attack, weapon	Criminology
4	flight, mistake, creates, helpful, focus, fight, outsmarted, impressively, excusing, situation	Aerodnetics
5	wort, retailer, independent, basis, healthiest, promotes, secret, drugnutrient, caught, spending	Thremmatology
6	pregnancy, pregnant, doula, contraction, labor, sperm, fertility, infertility, mother, midwife	Obstetrics
7	beauty, laughter, gender, aesthetic, male, hypnosis, female, ape, dolezal, mekong	Gender Studies

8	alzheimers, dementia, memory, cognitive, fatty, neuron, apoe, astrocyte, decline, aging	Neurology
9	lavender, smell, clean, cleaning, soap, laundry, scent, electrolyzed, rose, soda	Chemistry
10	bpa, amazon, bisphenol, canned, walmart, grocery, chain, yelp, clothing, retailer	Toxicology
11	rbc, vessel, oxygen, heat, epo, organ, temporary, fluid, guluzza, vein	Hematology
12	alamo, struck, skyrocketed, hide, mexico, debilitating, los, count, suddenly, deadly	Epidemiology
13	fighting, census, evidenced, youth, percentage, nations, physically, crime, alive, capable	Nomology
14	microorganism, microbiota, fly, dysbiosis, microbe, synonymous, microbiology, intestinal, prebiotics, intestine	Microbiology
15	weizmann, depriving, unleash, fasting, stating, mysterious, destroyed, blow, wonderful, altered	Nutrition
16	corn, south, acreage, farming, classification, fertilizer, total, price, bushel, land	Agronomy

17	curious, breaking, allowed	Natural Sciences
18	immense, handful, couple, paper, occur, eventually, probably, happened, apparently, jog	None
19	mozarts, volume, geometry, dialogue, sacred, volumetric, octahedron, timaeus, tetrahedron, hess	Musicology
20	particle, baryon, proton, electron, neutron, decay, atom, neutrino, meson, muon	Particle physics
21	pound, fitbit, lose, size, tangerine, tracker, losing, lost, fiona, fitness	Nutrition
22	religious, boswellia, popular, resin, potent, scent, traditional, offered, commonly, believed	Theology
23	shot, guessing, waiting, wrong, interject, incontrovertible, pressed, skeptic, insist, reject	Ethnomethodology
24	tropical, oleifera, continually, intricate, rosea, adaptogenic, warm, rhodiola, evergreen, exact	Botany

Note: Topic 18 was not assigned a topic name because its top 10 keywords did not clearly relate to any meaningful or coherent theme.

Discussion

The topics found in this study answer the research question by showing how junk science articles in the dataset really focus and cluster around a limited set of scientific areas rather than covering all the areas evenly. Based on the table above, this can be seen especially in health-related fields like neurology, hematology, microbiology, obstetrics, nutrition, and toxicology, also with agronomy and botany for agriculture. They also include social and theoretical disciplines like theology, gender studies, criminology, ethnomethodology, musicology, particle physics, and other natural sciences, showing that junk science often borrows labels from many branches of science have repeating themes more than others.

These results are useful because the labeled topics based on the table can make a large junk science corpus into structured overview that others can reuse for their own projects and further research. Misinformation researchers and system designers can use this table above to see which disciplines and issues are most frequently associated with junk science and focus on these areas when building or evaluating detection models.

This study also has limits. This study only uses one dataset, and the “junksci” label the author has given might sometimes be incorrect. The topic modeling model is automatic, so some topics may mix different ideas or include noise from data, and the final topic labels depend on personal judgement, as it is done manually.

Conclusion & Future Studies

The study used BERTopic to look at themes in 17,408 junk science articles from the Fake News Corpus. By cleaning the text and applying topic modeling, the study found 26 topics in total, including one outlier topic that mainly had random noise and 25 interpretable topics mapped to

different branches of science. with one of them being a random noise with the rest 25 being concrete topics. These topics showed that junk science articles in this dataset concentrate in certain areas, especially health-related fields (such as neurology, hematology, microbiology, obstetrics, nutrition, and toxicology), as well as agriculture, botany, theology, gender studies, criminology, musicology, particle physics, and other natural sciences.

Future work can be built on this study in several ways. One direction is to apply the same BERTopic approach to other misinformation categories in the Fake News Corpus, like conspiracy or political news, and compare their themes to junk science. Another direction is to study how these topics change over time or across different platforms. For example, comparing older and newer articles that share similar content can give different results. Finally, topic modeling can be combined with a classification model and can try to predict whether an article is likely to contain junk science. While there are other ways to study this further, few mentioned are just brainstormed ideas.

References

AIRLegend. (2025). *GitHub - AIRLegend/fakenews: Fake news detection using Deep Learning*.

GitHub. <https://github.com/AIRLegend/fakenews>

Arnfield, D. (2023). Enhanced Content-Based Fake News Detection Methods with Context-Labeled News Sources. *Electronic Theses and Dissertations*, 4269.

<https://dc.etsu.edu/etd/4269/>

Blowingfact. [@blowingfact]. (2025, December 1). Scientists in Japan can now erase memories like in a movie... . Instagram. <https://www.instagram.com/p/DRusfmdjpfE/>

Chung, H.-H., & Chen, J. (2025). Misinformation detection: datasets, models and performance.

Online Information Review, 49(3), 570–584. <https://doi.org/10.1108/oir-06-2024-0388>

Essa, E., Omar, K., & Alqahtani, A. (2023). Fake news detection based on a hybrid BERT and LightGBM models. *Complex Intell*, 9, 6581–6592. [https://doi.org/10.1007/s40747-](https://doi.org/10.1007/s40747-023-01098-0)

[023-01098-0](https://doi.org/10.1007/s40747-023-01098-0)

Grootendorst, M. (2022a). BERTopic: Neural topic modeling with a class-based TF-IDF

procedure. *ArXiv:2203.05794 [Cs]*, 1. <https://doi.org/10.48550/arXiv.2203.05794>

Grootendorst, M. (2022b). *Home - BERTopic*. Maartengr.github.io.

<https://maartengr.github.io/BERTopic/index.html>

Kumar, D. (2024, July 1). Topic Modelling with BERTopic. *Medium*. Topic Modelling with BERTopic

Matsui, K. (2024). *Saying Goodbye to Traumatic Memories: Astrocytic Manipulation of the Fate of Memory*. Tohoku University Official English Website.

https://www.tohoku.ac.jp/en/press/saying_goodbye_to_traumatic_memories_astrocyte_manipulation_of_the_fate_of_memory.html

PhD, M. N. (2023, August 21). Topic modeling algorithms. *Medium*.

<https://medium.com/@m.nath/topic-modeling-algorithms-b7f97cec6005>

Python Tutorials for Digital Humanities. (2022, September 16). *How to use BERTopic - Machine Learning Assisted Topic Modeling in Python*. YouTube.

<https://www.youtube.com/watch?v=v3SePt3fr9g>

several27. (2020, January 25). *GitHub - several27/FakeNewsCorpus: A dataset of millions of news articles scraped from a curated list of data sources*. GitHub.

<https://github.com/several27/FakeNewsCorpus/tree/master>

shribe! - master your studies. (2025, May 16). *Topic Modeling Explained (LDA, BERT, Machine Learning)* 🧠 📖 🔍. YouTube. <https://www.youtube.com/watch?v=-nqJ9Bzk8>